

List Comprehensions

Write clean, fast, Pythonic one-line lists

■ The Basic Idea

List Comprehensions are one of Python's absolute best features. They let you write clean, fast, and readable code by creating new lists in a single line — instead of writing a 4-line for loop with `.append()`, you can do the exact same thing in one clean line.

Let's look at how we used to create lists versus how we do it with list comprehension:

■ Old Way — for loop

```
numbers = [1, 2, 3, 4, 5]
squares = []

for num in numbers:
    squares.append(
        num * num
    )

print(squares)
# [1, 4, 9, 16, 25]
```

■ New Way — comprehension

```
numbers = [1, 2, 3, 4, 5]

squares = [
    num * num
    for num in numbers
]

print(squares)
# [1, 4, 9, 16, 25]
```

■ The Formula

List comprehension always follows this simple structure inside brackets `[...]`:

```
new_list = [ expression for item in iterable if condition ]
```

- **expression** — what you want to put in the new list (e.g., `num * num` or `name.upper()`).
- **for item in iterable** — the standard loop.
- **if condition** (optional) — filter items out (e.g., only keep even numbers).

■ Today's Practice Script: day73.py

Create a new file in VS Code: **backend/day73.py** and try these classic use cases:

Day 73 — List Comprehensions

```
# =====  
# DAY 73: LIST COMPREHENSIONS  
# =====  
  
# 1. Transform Data (Square numbers)  
numbers = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]  
squares = [x ** 2 for x in numbers]  
print("Squares:", squares)  
  
# 2. Filter Data (Only keep even numbers)  
evens = [x for x in numbers if x % 2 == 0]  
print("Even Numbers:", evens)  
  
# 3. Clean String Data (Backend use-case: format user input)  
raw_names = [" shoaib ", " ALI", " Ahmad "]  
clean_names = [name.strip().title() for name in raw_names]  
print("Clean Names:", clean_names)  
  
# 4. Combine Filtering & Transformation (Double only odd numbers)  
doubled_odds = [x * 2 for x in numbers if x % 2 != 0]  
print("Doubled Odds:", doubled_odds)
```

■ Your Task Today

- Type out backend/day73.py in VS Code.
- Run `python day73.py` in your terminal.
- Share your output here once you run it! ■

■ Day 73 Complete: Perfect Execution! ■

Your Terminal Output

```
Squares:          [1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
```

Day 73 — List Comprehensions

Even Numbers:	[2, 4, 6, 8, 10]
Clean Names:	['Shoaib', 'Ali', 'Ahmad']
Doubled Odds:	[2, 6, 10, 14, 18]

You mastered transforming data, filtering data with conditions, and cleaning up messy string lists — all in single-line comprehensions. This is a huge milestone for writing clean, Pythonic backend code!

■ Save & Push Day 73 to GitHub

Run these commands in your VS Code terminal to save your progress:

```
git add .
git commit -m "Day 73: Mastered list comprehensions for data transformation and filtering"
git push
```